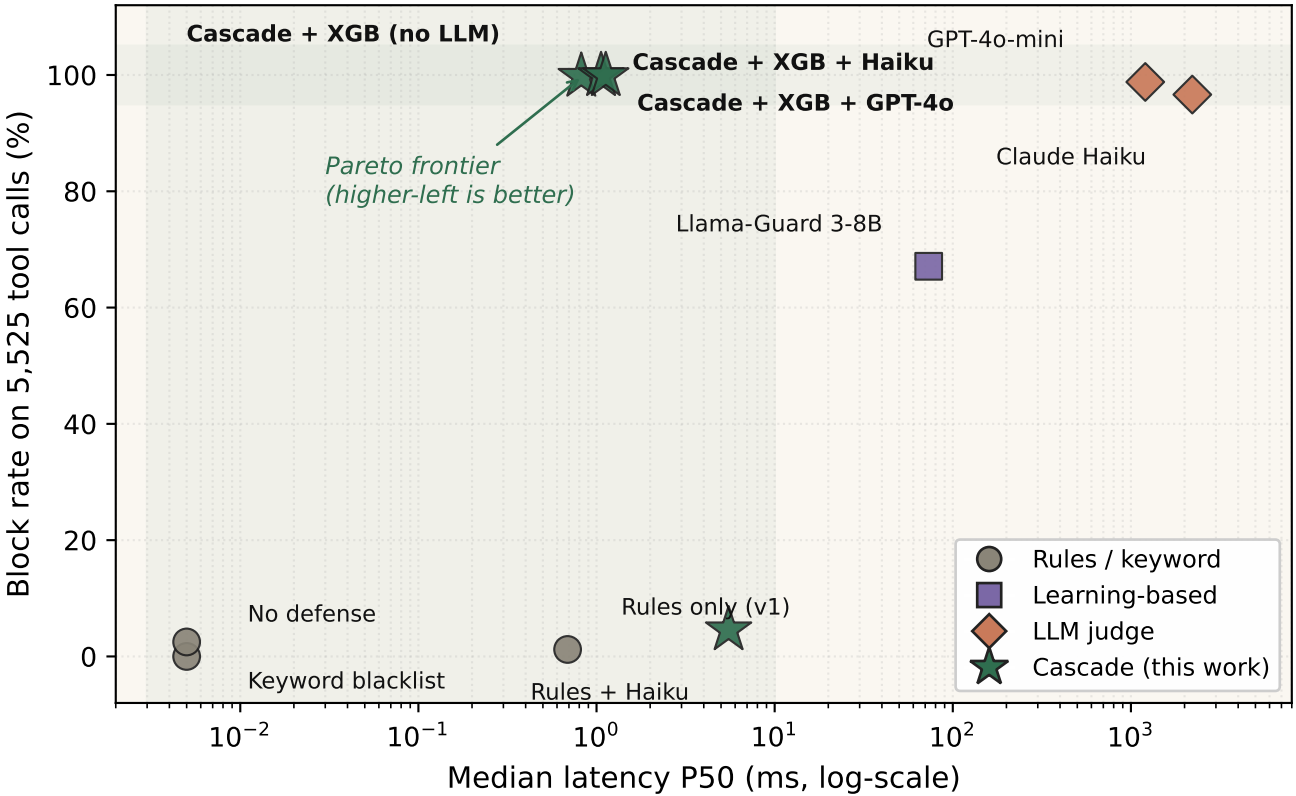
Cascade + XGB (no LLM)
Cascade + XGB + Haiku
Cascade + XGB + GPT-4o
GPT-4o-mini
Claude Haiku
Pareto frontier
(higher-left is better)
Llama-Guard 3-8B
No defense
Keyword blacklist
Rules only (v1)
Rules + Haiku
Rules / keyword
Learning-based
LLM judge
Cascade (this work)
Block rate on 5,525 tool calls (%)
Median latency P50 (ms, log-scale)
0
20
40
60
80
100
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}